

VITA: ZERO-SHOT VALUE FUNCTIONS VIA TEST-TIME ADAPTATION OF VISION-LANGUAGE MODELS

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ABSTRACT

Although Vision-Language Models (VLMs) exhibit considerable promise as zero-shot goal-conditioned value functions, the frozen pre-trained representations on which they depend circumscribe both their capacity for generalization and their competence in temporal reasoning. We introduce VITA, a method for zero-shot value function learning that ameliorates both deficiencies through test-time adaptation. At inference, a lightweight adaptation module is refined via a single gradient step on a meta-learned self-supervised objective, such that each test-time update demonstrably sharpens value estimation. Because these updates are applied sequentially across a trajectory, VITA progressively encodes the accumulated history into its parameters, thereby redressing the temporal-reasoning limitations that beset prior approaches. To forestall shortcut learning, we further devise a dissimilarity-based sampling strategy that, during training, selects semantically diverse segments of the trajectory. On real-world robotic manipulation tasks, VITA generalizes from a solitary training environment to a heterogeneous array of out-of-distribution tasks, environments, and embodiments, thereby surpassing the state-of-the-art zero-shot method predicated on autoregressive VLMs. Beyond this, we establish that the zero-shot value estimates furnished by VITA lend themselves to reward shaping in offline reinforcement learning, yielding multi-task policies on the Meta-World benchmark that eclipse the performance of those trained with the simulation’s fuzzy-logic dense rewards.

1 INTRODUCTION

Vision-Language Models (VLMs), by learning from large-scale, unstructured web corpora in the absence of human supervision, have evinced a robust capacity for generalization that spans heterogeneous tasks and domains (Radford et al., 2021; Alayrac et al., 2022). By contrast, and notwithstanding the considerable strides made in the acquisition of generalist policies for robotics (Brohan et al., 2023; Kim et al., 2024) and for 3D virtual environments (Reed et al., 2022; SIMA Team et al., 2024), state-of-the-art methods have yet to attain a commensurate degree of success, an outcome attributable in large measure to their dependence on expert demonstrations (Ho & Ermon, 2016). World models (Ha & Schmidhuber, 2018; Hafner et al., 2021; Matsuo et al., 2022) have accordingly emerged as an auspicious remedy, endowing agents with the ability to learn within simulated environments, and their applications now extend across robotics (Yang et al., 2023; Agarwal et al., 2025), autonomous driving (Hu et al., 2024), and video games (Bruce et al., 2024). Yet even granting the capacity of world models to synthesize realistic visual trajectories in diverse, open-ended environments (Hughes et al., 2024; Silver & Sutton, 2025), one pivotal challenge persists: *how can agents learn effectively from videos at scale without recourse to human supervision?*

One line of inquiry pursues the acquisition of policies directly from expert visual trajectories, inferring the underlying actions from the observed visual transitions (Edwards et al., 2019; Baker et al., 2022; Schmidt & Jiang, 2024; Zhang et al., 2025a). Although such latent actions can, at deployment, be mapped onto executable controls, this mapping remains a formidable undertaking in practice. A parallel body of work concentrates instead on learning a universal *goal-conditioned value function*, subsequently repurposing it as a zero-shot reward-shaping and supervision signal for reinforcement learning (RL) and imitation learning, respectively (Chen et al., 2021; Ma et al., 2022). Within this framework, goal-conditioned value estimation admits a natural formulation: predicting the extent to which an agent has advanced toward the completion of a task, conditioned jointly on its visual

observations and a natural-language task description (Lee et al., 2021a; Ma et al., 2023b; 2024; Dashora et al., 2025).

Prior work has recruited pre-trained contrastive VLMs (Radford et al., 2021) for zero-shot reward shaping, exploiting the affinity between task descriptions and visual observations within a shared multimodal representation space so as to anchor progress in semantic context (Ma et al., 2023b; Baumli et al., 2023; Rocamonde et al., 2024). Such methods, however, prove incapable of apprehending the temporal context requisite for disambiguating visually congruent states that recur at disparate stages of a task (for instance, the folding as against the unfolding of a shirt). Autoregressive VLMs (Alayrac et al., 2022), by contrast, assimilate temporal context by conditioning on the entire visual trajectory embedded within the prompt; yet they inherit, from the chronologically ordered datasets on which they are pre-trained, a pronounced predisposition toward monotonically increasing predictions (Ma et al., 2024). Both architectures, in the end, lean upon pre-trained representations (Ma et al., 2023b; 2024; Baumli et al., 2023; Rocamonde et al., 2024; Du et al., 2023) for zero-shot prediction, a reliance that constrains their generalization and their temporal reasoning alike (Pătrăucean et al., 2023). In pursuit of generalizable value function learning, recent work has canvassed large-scale pre-training (Alakuijala et al., 2024; Ma et al., 2023b), domain-specific fine-tuning (Zhang et al., 2025b), and demonstration-based reward specification (Sontakke et al., 2023).

In the present work, we introduce VITA, a method for zero-shot value function learning that, by means of test-time adaptation, augments both generalization and temporal reasoning, and that thereby outperforms the state-of-the-art zero-shot approach (Ma et al., 2024) on real-world robotic manipulation tasks. In contradistinction to prior methods, which predicate their generalization upon large-scale pre-training or expert demonstrations, VITA adapts online to the semantic and temporal context of trajectories alike. At inference, a lightweight adaptation module is updated in negligible time through a gradient step on a meta-learned self-supervised loss (Sun et al., 2024), so contrived that each test-time update improves value function estimation (Finn et al., 2017). By applying these updates sequentially over a test trajectory, the value function estimator inscribes the trajectory history into its own parameters (Sun et al., 2024), and in so doing redresses the temporal-reasoning limitations that have encumbered prior work. To attenuate shortcut learning (Geirhos et al., 2020), we further introduce a dissimilarity-based sampling strategy that, as our empirical evidence attests, encourages reliance on semantic cues. Our evaluation substantiates the efficacy of the method across the core capabilities of value function estimation: generalization under distribution shifts, the differentiation of expert from non-expert trajectories, and reward shaping for offline RL (Levine et al., 2020).

We summarize our key contributions as follows:

- We propose VITA, a test-time adaptation method that augments the generalization and temporal reasoning of contrastive VLMs for zero-shot value function estimation, dispensing altogether with task-specific demonstrations and large-scale pre-training.
- VITA generalizes from a solitary training environment to a diverse assortment of out-of-distribution tasks, environments, and embodiments in robotic manipulation, thereby outperforming the state-of-the-art zero-shot method (Ma et al., 2024).
- When marshalled for reward shaping, the zero-shot value estimates furnished by VITA yield offline RL policies on the Meta-World benchmark for multi-task learning (MT10) that surpass those trained with the simulation’s fuzzy-logic dense rewards (Yu et al., 2020).

2 PRELIMINARIES

2.1 GOAL-CONDITIONED VALUE FUNCTIONS

Within video trajectories, the estimation of task progress may be construed as goal-conditioned value function estimation, wherein the reward is taken to reflect task completion. Accordingly, we cast the learning of a goal-conditioned value function as the prediction, from vision–language representations, of the degree to which a task has been completed. Formally, the value function is defined as $V : \mathcal{O} \times \mathcal{G} \rightarrow [0, 1]$, a mapping that carries an observation $o_t \in \mathcal{O}$ and a goal specification $g \in \mathcal{G}$ to a scalar quantifying the predicted progress toward goal completion. By convention, we let $V(o_t; g) = 0$ denote the inception of the task and $V(o_t; g) = 1$ its consummation. Task progress is customarily placed in correspondence with temporal position in expert demonstrations, on the supposition that

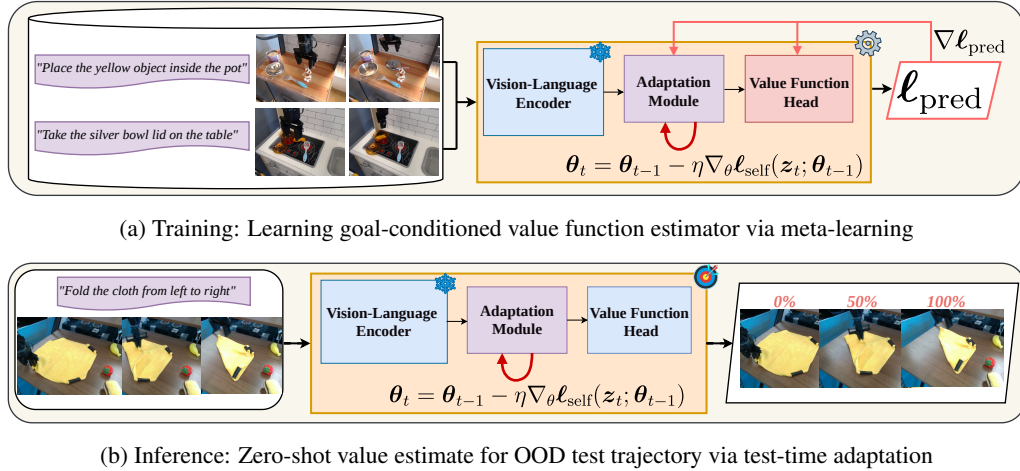


Figure 1: **Overview of VITA.** VITA learns a goal-conditioned value function via meta-learning and achieves zero-shot generalization to out-of-distribution trajectories via test-time adaptation.

such trajectories manifest monotonically increasing progress toward goal completion (Lee et al., 2021a; Ma et al., 2024; Dashora et al., 2025). Given an expert trajectory $\tau = (o_1, \dots, o_T) \sim \pi_E$, temporal progress is defined by way of normalized timestep indices: $V^{\pi_E}(o_t; g) = \frac{t}{T}$, where T denotes the trajectory length. Temporal progress thus furnishes the supervisory signal for learning a goal-conditioned value function V .

2.2 TEST-TIME TRAINING FOR SEQUENCE MODELING

Test-time training (TTT) (Sun et al., 2020) constitutes a test-time adaptation method that reconceives inference as a self-supervised learning task (Goodfellow et al., 2016), updating the model’s parameters on each test instance in the absence of any labels. Although originally conceived for static image classification (Sun et al., 2020), TTT has more recently been extended to sequence modeling (Sun et al., 2024; Wang et al., 2023), wherein a parametric adaptation module f_{adapt} is updated at each timestep through a gradient step on a self-supervised loss. Whereas conventional sequence architectures lodge temporal history within activation states—be they hidden states (Hochreiter & Schmidhuber, 1997; Cho et al., 2014) or cached key–value pairs (Vaswani et al., 2017)—TTT instead inscribes that history into the parameters of f_{adapt} by means of sequential updates carried out at inference. The parameters of f_{adapt} consequently operate as an implicit memory, one that encodes temporal history while faithfully preserving temporal order. Sun et al. (2024) go further, proposing to meta-learn the self-supervised task in accordance with the gradient-based meta-learning paradigm (Finn et al., 2017). In that setting (Sun et al., 2024), the self-supervised loss is parameterized by learnable linear projections and trained not to minimize the downstream prediction loss directly, but rather to minimize it subsequent to a test-time update. In the present work, we bring TTT to bear on goal-conditioned value function learning, whereby temporal context is captured through sequential updates and the self-supervised task is meta-learned so as to improve value estimation, in lieu of being stipulated a priori.

3 VITA: ZERO-SHOT VALUE FUNCTIONS VIA TEST-TIME ADAPTATION

3.1 MODEL ARCHITECTURE

Our goal-conditioned value function estimator is constituted of three modules: (1) a multimodal encoder that extracts joint visual–language representations from visual trajectories together with their task descriptions; (2) an adaptation module that is updated at test time by means of a self-supervised loss meta-learned so as to improve value estimation; and (3) a regression head that emits the value estimates themselves.

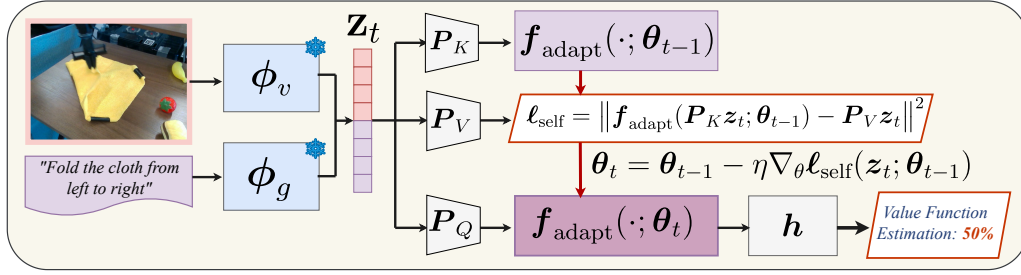


Figure 2: **Test-time Adaptation.** At inference, and at every timestep t , an adaptation module f_{adapt} is updated through a gradient step on a meta-learned self-supervised loss ℓ_{self} , thereby encoding the temporal history.

3.1.1 MULTIMODAL INPUT REPRESENTATION

We employ a frozen contrastive vision–language encoder, CLIP (Radford et al., 2021), to derive representations from both the visual observations and the goal descriptions. Given a visual trajectory $\tau = (o_1, \dots, o_T)$ and a linguistic task description g , we concatenate their respective representations at each timestep t , thereby assembling a sequence of joint multimodal representations $(z_1, z_2, \dots, z_T)$, where $z_t = [\phi_v(o_t); \phi_g(g)] \in \mathbb{R}^{2d}$, and where ϕ_v and ϕ_g denote the visual and language encoders, respectively. Inasmuch as CLIP is pre-trained with a contrastive objective that aligns paired image–text inputs within a shared representation space (Radford et al., 2021), the representations of visual observations that lie semantically nearer to goal completion tend, in turn, to lie nearer to the representation of the goal description.

3.1.2 TEST-TIME ADAPTATION

In order to attune the multimodal representations to the semantic and temporal context alike, we deploy an adaptation module f_{adapt} that conforms to the test-time training paradigm delineated in 2.2. At inference, the parameters of f_{adapt} are updated online through a gradient step on a meta-learned self-supervised loss ℓ_{self} . This loss is cast as a reconstruction objective mediated by learnable linear projections P_K and P_V , themselves meta-learned during training so as to improve value estimation following test-time adaptation. Specifically, given $P_K \in \mathbb{R}^{d' \times 2d}$, which engenders a perturbed view of the input, and $P_V \in \mathbb{R}^{d' \times 2d}$, which supplies the target, the self-supervised loss is defined as:

$$\ell_{\text{self}}(z_t; \theta_{t-1}, P_K, P_V) = \|f_{\text{adapt}}(P_K z_t; \theta_{t-1}) - P_V z_t\|^2. \quad (1)$$

where θ_{t-1} denotes the parameters of f_{adapt} before test-time adaptation. At each timestep t , the parameters are updated by

$$\theta_t = \theta_{t-1} - \eta \nabla_{\theta} \ell_{\text{self}}(z_t; \theta_{t-1}), \quad (2)$$

with learning rate η . By virtue of these sequential updates, f_{adapt} implicitly retains information gleaned from past visual observations, and thereby captures the temporal context. In Section 4.6.2, we adduce empirical evidence that incremental, parameter-level updates of the value function estimator are more efficacious than updates enacted at the level of the entire trajectory.

3.1.3 ZERO-SHOT VALUE FUNCTION ESTIMATOR

Once test-time adaptation has been carried out, a meta-learned linear projection $P_Q \in \mathbb{R}^{d' \times 2d}$ maps the input z_t into an adaptation space $\mathbb{R}^{d'}$. This representation is then routed through the adaptation module f_{adapt} , and thereafter through a regression head h that emits the predicted values of the goal-conditioned value function:

$$V(z_t; g) = h(f_{\text{adapt}}(P_Q z_t; \theta_t)) \quad (3)$$

The regression head h takes the form of a two-layer multilayer perceptron (MLP), in keeping with the value-function estimator architectures customarily employed in deep reinforcement learning (Espeholt

et al., 2018). The network is trained on the normalized progress labels y_t drawn from expert demonstrations as described in Section 2.1, under a supervised prediction loss ℓ_{pred} defined as the mean squared error between the predicted values $V(z_t; g)$ and the targets y_t . At inference, the regression head h is held frozen, while the adaptation module f_{adapt} is updated online under the self-supervised objective—an arrangement that equips the value function estimator to generalize in a zero-shot regime without recourse to task-specific demonstrations.

3.2 TRAINING PROCEDURE

We train the value function estimator by means of gradient-based meta-learning, optimizing a self-supervised loss ℓ_{self} in such a manner that test-time adaptation improves the supervised prediction loss ℓ_{pred} (Sun et al., 2024). Training proceeds over diverse sub-trajectories culled via dissimilarity-based sampling, a procedure that at once attenuates shortcut learning and fosters reliance on semantic and temporal cues.

3.2.1 META-LEARNING

During training, the adaptation module f_{adapt} is updated online at each timestep under the self-supervised loss ℓ_{self} . In accordance with the gradient-based meta-learning paradigm (Finn et al., 2017), we differentiate through these adaptation updates with respect to the supervised prediction loss ℓ_{pred} , thereby optimizing the initialization θ_0 of f_{adapt} , the linear projections P_K, P_V, P_Q , and the regression head h in concert. Formally, the total training loss conjoins the supervised prediction loss ℓ_{pred} with the self-supervised loss ℓ_{self} , the latter weighted by a scalar λ . The learned parameters θ_0 subsequently serve as the initialization for adaptation at inference. This scheme endows the value function estimator with the capacity to learn how to adapt its internal representations at test time, and so to sharpen value function estimation.

3.2.2 DISSIMILARITY-BASED SAMPLING

Expert visual trajectories frequently harbor consecutive frames of pronounced redundancy, a phenomenon noted in prior work on video action recognition (Wang et al., 2018). Such redundancy is liable to induce the value function estimator to capitalize on shortcut cues (Geirhos et al., 2020; Lee et al., 2021b), overfitting to the late-stage visual patterns that recur with high frequency. To counteract this tendency, we propose a dissimilarity-based sampling strategy that assembles mini-batches from the most visually diverse sub-trajectories within each trajectory. In so doing, it amplifies intra-batch variance and operates as a species of importance sampling that privileges semantic diversity.

Formally, given a sequence of multimodal representations $\{z_1, \dots, z_T\}$, we extract fixed-length sub-trajectories by sliding a window of size w_{tr} with stride s , which yields a candidate set $\mathcal{W}$. A diverse subset of size k is then taken to be one that maximizes pairwise dissimilarity:

$$\mathcal{W}' = \arg \max_{\mathcal{U} \subset \mathcal{W}, |\mathcal{U}|=k} \sum_{\{w^i, w^j\} \in \binom{\mathcal{U}}{2}} \|w^i - w^j\|_2^2, \quad (4)$$

where each w^i constitutes a sub-trajectory of length w_{tr} . Solving Eq. 4 exactly is intractable, inasmuch as it would demand the enumeration of all $\binom{|\mathcal{W}|}{k}$ subsets. To approximate this objective efficiently, we therefore adopt a scoring-based heuristic that assigns to each window $w \in \mathcal{W}$ a score equal to its aggregate dissimilarity from all others:

$$s(w) = \sum_{v \in \mathcal{W}} \|w - v\|_2^2, \quad \mathcal{W}' = \arg \max_{\mathcal{U} \subset \mathcal{W}, |\mathcal{U}|=k} \sum_{w \in \mathcal{U}} s(w). \quad (5)$$

Thus, by computing a diversity score s for each window and retaining the k highest-scoring windows, we obtain a heuristically diverse subset at polynomial-time cost. As our complexity analysis in Appendix G establishes, the resultant overhead is negligible relative to that of model training. We validate dissimilarity-based sampling empirically against full-trajectory sampling in the ablation study (Section 4.6.1), where it exhibits an enhanced capacity to distinguish expert from non-expert visual trajectories.

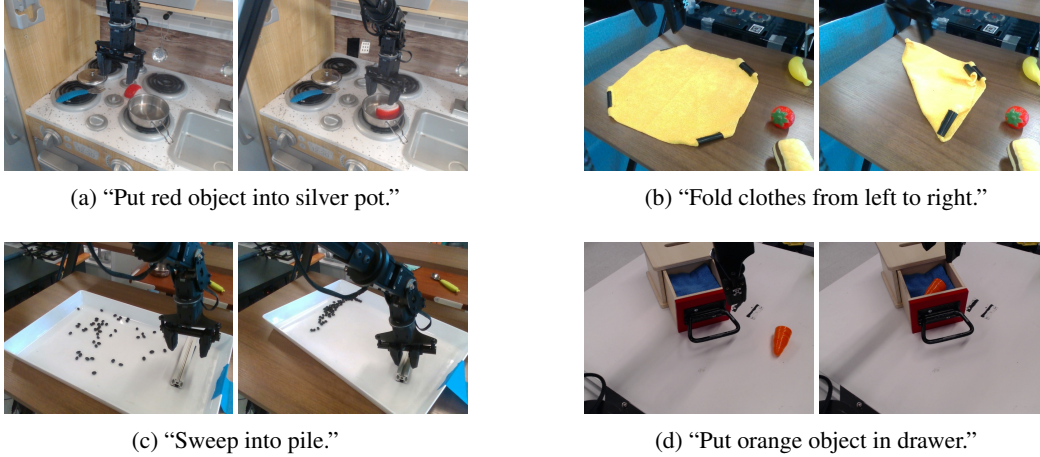


Figure 3: Representative visual trajectories, each paired with its task description, exhibited under a range of distribution shifts. (a) In-distribution. (b, c) Environment shift. (d) Conjoint embodiment and environment shift.

4 EXPERIMENTS

Our evaluation attests to the efficacy of VITA across the core capabilities that a goal-conditioned value function must possess: (i) generalization in value estimation for real-world robotic manipulation under distribution shifts in task, environment, and embodiment; (ii) the differentiation of expert from non-expert visual trajectories in real-world robotic manipulation tasks; and (iii) effective zero-shot reward shaping for offline RL on Meta-World MT10, a simulated benchmark for multi-task robot learning.

4.1 TRAINING SETUP

We train VITA on expert visual trajectories paired with natural-language task descriptions drawn from the BridgeData V2 dataset (Walke et al., 2023), which encompasses a broad gamut of manipulation tasks, environments, and robot embodiments. Concretely, we draw upon a curated subset comprising 2,986 expert demonstrations that cover pick-and-place manipulation tasks, yet deliberately exclude folding, sweeping, and stacking. The whole of these demonstrations was collected with a single robot embodiment (WidowX 250) across 4 configurations of the ToyKitchen environment. A further 287 expert demonstrations are sequestered as the in-distribution test set, which we designate $\tau_{k\text{-pnp}}$; representative examples from the training set appear in Appendix A. As the frozen backbone for encoding video frames and task descriptions, we adopt OpenCLIP ViT-B/32. The training objective conjoins ℓ_{pred} with ℓ_{self} , the latter weighted by $\lambda_{\text{self}} = 0.5$. Throughout training, we employ dissimilarity-based sampling with window size $w_{\text{tr}} = 8$, a sub-trajectory count of $k = 8$, and stride $s = 1$. At test time, we perform a single gradient step ($t_{\text{ep}} = 1$) under λ_{self} with learning rate $\eta = 0.1$. Further particulars, together with the attendant hyperparameter sweeps, are reported in Appendix D.

4.2 BASELINES

We benchmark VITA against value functions predicated on contrastive and autoregressive VLMs alike. VLM-CL derives its zero-shot value estimates from the cosine similarity between the CLIP representations of each frame and those of the task description (Baumli et al., 2023). VLM-RM regularizes the CLIP embeddings by projecting the features along the direction leading from a generic reference prompt to the task prompt (Rocamonde et al., 2024). CLIP-FT trains a supervised regression head atop frozen CLIP multimodal representations, whereas CLIP-GRU trains a Gated Recurrent Unit (GRU) (Cho et al., 2014) on the same frozen CLIP multimodal representations. GVL (Ma et al., 2024) constitutes the state-of-the-art zero-shot value function, one that harnesses autoregressive VLMs. In our experiments, GVL-0S designates the zero-shot setting, in which the VLM is furnished with the test trajectory and task description alone, while GVL-1S corresponds to a

one-shot in-context setting, in which a full shuffled trajectory drawn from the training distribution, accompanied by its progress labels, is supplied as an exemplar. In common with GVL, we employ Gemini 1.5 Pro (gemini-1.5-pro-latest) (Gemini Team et al., 2024) together with its proposed prompt template. We additionally examined Qwen-VL 2.5 (Yang et al., 2024), which proved unable to surmount the temporal bias, and GPT-4o (Hurst et al., 2024), which declined altogether to produce value estimates. Supplementary implementation details are provided in Appendix D.

4.3 EVALUATING GENERALIZATION UNDER DISTRIBUTION SHIFTS

Table 1: VOC scores for value function estimation under an assortment of distribution shifts. **ID** = In-Distribution, **ES** = Environment Shift, **EM** = Embodiment Shift, **ES & EM** = Conjoint Shifts.

Shift	Dataset	VLM-CL	VLM-RM	CLIP-FT	GVL-0S	GVL-1S	CLIP-GRU	VITA
ID	tk_pnp	0.038	0.029	0.251	0.269	0.252	<u>0.773</u>	0.782
ES	lm_pnp	0.017	0.033	0.149	0.305	0.272	<u>0.676</u>	0.725
	td_fold	0.031	0.072	0.152	0.326	0.318	<u>0.674</u>	0.709
	ft_fold	0.108	0.099	0.162	0.331	0.387	0.693	<u>0.658</u>
	rd_fold	0.095	0.055	0.126	0.372	0.406	0.726	<u>0.606</u>
	ms_sweep	-0.129	-0.226	0.148	0.158	0.150	<u>0.434</u>	0.490
EM	dt_tk_pnp	0.042	-0.041	0.149	0.258	0.211	0.856	<u>0.820</u>
	dt_tk_stack	0.035	0.046	0.099	0.254	0.277	<u>0.667</u>	0.708
ES & EM	dt_ft_stack	0.026	0.028	0.049	0.212	0.249	<u>0.674</u>	0.698
	dt_rd_pnp	0.023	0.041	0.211	0.329	0.316	0.747	<u>0.695</u>

We assess the capacity of progress estimators to generalize across novel environments, tasks, and robot embodiments by drawing upon subsets of BridgeData V2 that were curated expressly to introduce variation along each of these axes. Environment shifts entail alterations of scene layout or background: `lm_pnp`, for instance, is a pick-and-place task situated before a laundry machine, whereas `td_fold`, `ft_fold`, and `rd_fold` involve cloth folding across disparate surfaces, and `ms_sweep` introduces a long-horizon sweeping task confined to a tray. Embodiment shifts, in turn, are appraised by recourse to the DeepThought robot, which diverges from the training embodiment (WidowX 250) in both morphology and camera perspective. Whereas `dt_tk_pnp` (pick-and-place) and `dt_tk_stack` (stacking) preserve the in-distribution environment while introducing a novel embodiment, `dt_ft_stack` (stacking) and `dt_rd_pnp` (drawer pick-and-place) implicate embodiment and environment shifts in tandem. Each dataset comprises 200 expert trajectories with task descriptions, the sole exception being `ms_sweep`, which, owing to its limited size, contains 100. A full description of our evaluation datasets is furnished in Appendix B. We gauge the performance of a value function estimator by means of the *Value Order Correlation* (VOC) metric (Ma et al., 2024), which quantifies the concordance between the predicted progress toward task completion and the chronological ordering of the frames within a visual trajectory. Formally, let $p_1, \dots, p_T$ denote the predicted progress values for a trajectory of T frames, and let $r_t = t$ represent the temporal index; VOC is then defined as the Spearman rank correlation ρ_s between the predicted values and the temporal indices.

The results reported in Table 1 reveal that both VITA and CLIP-GRU consistently outperform GVL-0S and GVL-1S, a finding that at once lays bare the limitations of pre-trained autoregressive VLMs and underscores the salience of preserving temporal order for effective progress estimation. Although both estimate progress effectively, VITA prevails over CLIP-GRU on the preponderance of the evaluation datasets (6 of 10). Both VITA and CLIP-GRU, moreover, outperform the CLIP-based baselines that forgo temporal modeling (CLIP-FT, VLM-CL, VLM-RM), a contrast that attests to the consequence of encoding temporal history for progress estimation. CLIP-FT performs on a par with GVL on the in-distribution task, yet fails to generalize to out-of-distribution settings; VLM-CL and VLM-RM alike perform poorly, in all likelihood owing to their want of temporal modeling. As for GVL-0S and GVL-1S, in-context examples prove salutary in certain cases while, in others, conferring no benefit or even precipitating degradation. Both GVL methods acquit themselves well on the folding tasks (`td_fold`, `ft_fold`, `rd_fold`), but fall short of comparable performance on the stacking (`dt_tk_stack`, `dt_ft_stack`) and pick-and-place (`dt_tk_pnp`, `lm_pnp`) tasks, an

asymmetry that intimates a bias, on the part of the autoregressive VLM underlying GVL, toward folding-like robotic manipulations. VITA, by contrast, attains consistent performance across all task types and distribution shifts, thereby evincing a generalization more robust than that of in-context learning.

On the long-horizon task `ms_sweep`, every method save CLIP-FT suffers a decline in performance, yet VITA nonetheless secures the highest VOC score. It outperforms CLIP-GRU in particular, which suggests that, for long-horizon robotic manipulation tasks, encoding temporal history via test-time adaptation is more efficacious than modeling it through recurrent hidden states. Under embodiment shifts, VITA likewise exhibits strong generalization. On `dt_tk_pnp`, which pairs a distinct robot embodiment with the same environment and tasks as the training set, both VITA and CLIP-GRU surpass their own in-distribution performance—an indication that the learning of temporal context can transfer effectively across robot embodiments. GVL-1S, by contrast, performs poorly on both `dt_tk_pnp` and `dt_tk_stack`, which intimates that few-shot learning fails to generalize under embodiment shifts within our evaluation setup.

4.4 DIFFERENTIATING EXPERT FROM NON-EXPERT TRAJECTORIES

Beyond generalization, we probe the robustness of VITA by examining its ability to discriminate expert from suboptimal visual trajectories, assigning to the latter the lower progress scores they warrant. To this end, we juxtapose the model’s predictions on expert and scripted (non-expert) visual trajectories collected from the same in-distribution setting of BridgeData V2. The scripted trajectories are generated within the ToyKitchen environment by a heavily randomized controller (Walke et al., 2023), under the very embodiment and environmental configuration that gave rise to the expert demonstrations. They comprise pick-and-place tasks that implicate object categories liable to overlap with the training set—namely, general objects (`sc_pnp_obj`), rigid items (`sc_pnp_obj`), soft toys (`sc_pnp_stoy`), utensils (`sc_pnp_uten`), and a cloth (`sc_pnp_cloth`). Although these trajectories accord with the training distribution in task, embodiment, and environment, their behavior departs from the smooth and monotonic progress that expert demonstrations typically display. Examples from the scripted dataset appear in Appendix C. Since the scripted datasets issue from a randomized controller, they are not contrived to reflect quantifiable gradations of suboptimality; an effective value function estimator should nevertheless, when evaluated in the same setting, ascribe lower VOC scores to suboptimal trajectories than to expert ones. We quantify discriminative success by means of *BinVOC*, a binary evaluation metric defined as $1 [\text{VOC}_{\text{exp}} > \text{VOC}_{\text{subopt}}]$, which assumes the value 1 when expert trajectories attain a higher VOC than suboptimal ones and 0 otherwise.

The discriminative performance of the model is assessed via *BINVOC*, averaged over the totality of the scripted datasets, while its performance on expert visual trajectories (VOC_{exp}) is measured on the in-distribution test set `tk_pnp`, which comprises pick-and-place tasks kindred to those of the scripted evaluation. VITA, GVL-0S, and GVL-1S each achieve flawless discrimination, invariably conferring higher VOC scores upon expert trajectories across all tasks. That VITA outperforms CLIP-GRU intimates that the encoding of temporal memory within recurrent hidden states may be more susceptible to overfitting to temporal shortcuts than is the implicit memory afforded by sequential test-time updates. The CLIP-based baselines (VLM-CL, VLM-RM) presumably underperform for want of temporal modeling, whereas CLIP-FT performs more dependably yet falters on a single task, betraying a sensitivity to object distribution shifts.

4.5 ZERO-SHOT REWARD SHAPING FOR OFFLINE RL

We evaluate our value function estimator on the Meta-World MT10 benchmark (Yu et al., 2020), which is composed of ten diverse robotic manipulation tasks—among them pick-and-place, door opening, and pushing. This experiment is devised expressly to appraise the zero-shot reward shaping that VITA affords for offline RL: VITA is trained on real-world robotic data and then evaluated zero-shot within this simulated setting. For each task of the Meta-World MT10 benchmark, we generate 20 expert visual demonstrations by way of Meta-World’s expert policies, whereupon the goal-conditioned value estimator is enlisted to define a dense reward for each visual observation. Policies are trained with Implicit Q-Learning (IQL) (Kostrikov et al., 2021) in the offline RL regime, and we replicate both the expert data generation and the policy training across 10 random seeds. Upon completion of training, each multi-task policy is evaluated over 20 rollouts per task, and we

Method	BinVOC
VLM-CL	0.40
VLM-RM	0.00
CLIP-FT	0.80
GVL-OS	1.00
GVL-1S	1.00
CLIP-GRU	0.80
VITA	1.00

(a) Expert vs. Non-Expert (BinVOC).

Method	IQM	95% CI
VLM-CL	0.760	[0.722, 0.791]
VLM-RM	0.746	[0.718, 0.771]
CLIP-FT	0.785	[0.759, 0.809]
CLIP-GRU	0.777	[0.734, 0.814]
META-WL	0.779	[0.750, 0.804]
VITA	0.815	[0.785, 0.838]

(b) Offline RL on MT10 (IQM).

Table 2: (a) Success in discriminating expert from scripted robot demonstrations, as measured by the average BinVOC across 5 in-distribution scripted datasets. (b) Offline RL performance on the Meta-World MT10 benchmark, as measured by the interquartile mean (IQM) together with 95% stratified bootstrap CIs.

compute the average return across episodes for each (task, seed) pair. Following the evaluation protocol advanced by Agarwal et al. (2021), we report the interquartile mean (IQM) across all pairs, accompanied by 95% stratified bootstrap confidence intervals. The training details and the evaluation protocol are elaborated further in Appendix E.

Table 2 sets out the performance of the baselines on the MT10 benchmark under the evaluation protocol described above. VITA attains the highest IQM return (0.815), thereby outperforming every CLIP-based baseline. A direct comparison with GVL is infeasible at scale, inasmuch as it depends on Gemini-1.5, a proprietary autoregressive VLM whose inference cost proves prohibitive in RL settings, while the open-source alternatives yielded no reliable progress estimates in our preliminary tests. We further report the fuzzy-logic dense reward supplied by Meta-World (META-WL), which reaches 0.779. Notwithstanding its considerable strength, META-WL is surpassed by VITA—a result that testifies to the capacity of a value estimator trained on real-world data to generalize effectively to simulated reward shaping.

4.6 ABLATION STUDIES

4.6.1 EFFECT OF DISSIMILARITY-BASED SAMPLING ON DISCRIMINATIVE PERFORMANCE

We validate the efficacy of the proposed dissimilarity-based sampling empirically, contrasting the effect of divergent sub-trajectory sampling strategies upon the ability of VITA to distinguish expert from non-expert demonstrations. Full-trajectory sampling (Sun et al., 2024), for its part, computes the adaptation loss over all overlapping sub-trajectories with stride $s = 1$, a practice that, in our setup, precipitates overfitting to global temporal shortcuts. Random sampling ameliorates diversity by selecting sub-trajectories uniformly, but remains bereft of semantic diversity. Dissimilarity-based sampling, by contrast, deliberately cultivates semantic diversity and thereby yields superior generalization: our proposed method, which selects those sub-trajectories that maximize pairwise dissimilarity, outperforms both full-trajectory and random sampling in the differentiation of expert from non-expert visual trajectories. For further particulars, we refer the reader to Appendix F.1.

4.6.2 EFFECT OF TEMPORAL MEMORY ON VALUE FUNCTION ESTIMATION

We analyze how divergent test-time adaptation strategies for the incorporation of temporal context bear upon value function estimation. VITA performs step-by-step adaptation without reset (implicit memory), incrementally updating its parameters from θ_{t-1} ; this permits the value function estimator to inscribe the trajectory history directly into its parameters, capturing at once the temporal order and the accumulated temporal context. We set VITA against three alternatives: (i) *trajectory-level adaptation (TTT-TR)*, which enacts a single gradient update computed over the full trajectory, and which, in producing a batched update that averages the loss across all frames, thereby discards temporal order; (ii) *memoryless adaptation (TTT-RS)*, which resets the adaptation module to θ_0 at every step and, in adapting solely to the current visual observation, forecloses the propagation of any temporal information; (iii) *explicit memory (TTT-EX)*, which likewise resets the adaptation module at every step but updates it with a local window of recent frames matching the window size used

during meta-training ($w_{tr} = 8$). Across the whole of the datasets, VITA consistently outperforms the trajectory-level, memoryless, and explicit-memory methods, thereby demonstrating the efficacy of the incremental accumulation of temporal context for value function estimation. The full analysis is provided in Appendix F.2.

5 RELATED WORK

5.1 VLMS AS VALUE FUNCTION ESTIMATORS

Contrastive VLMS have been pressed into service as zero-shot reward models (Baumli et al., 2023; Rocamonde et al., 2024) and as goal-conditioned value functions (Ma et al., 2023b; 2024), relying on frame-level similarity scores yet lacking any temporal modeling. Certain works fine-tune CLIP on domain-specific video datasets (Fan et al., 2022; Jiang et al., 2024) in order to generate dense rewards, thereby incurring a supervisory burden that curtails their zero-shot applicability. RoboCLIP (Sontakke et al., 2023) makes possible one-shot reward specification without fine-tuning, but it yields sparse rewards and continues to presuppose demonstrations. Large-scale multimodal pre-training approaches (Alakuijala et al., 2024; Li et al., 2024; Ma et al., 2023b) learn language-conditioned value functions by staking their generalization upon sheer scale, while ReWiND (Zhang et al., 2025b) trains a cross-modal sequential transformer on augmented large-scale data through video rewinding, conjoined with a handful of in-domain demonstrations, so as to bolster generalization. VITA, by contrast, adapts a pre-trained contrastive vision–language encoder at inference, apprehending both temporal and semantic context through test-time adaptation and dispensing with the large-scale multimodal pre-training or supplementary domain-specific demonstrations that generalization would otherwise demand. VITA could, moreover, be brought to bear on any pre-trained vision–language representation encoding a universal value function, so as to augment its generalization and temporal reasoning by way of meta-learned test-time adaptation. Autoregressive VLMS (Alayrac et al., 2022), which capitalize on in-context learning over trajectories, have been applied to success detection (Du et al., 2023) and to goal-conditioned value functions (Ma et al., 2024). GVL (Ma et al., 2024) tempers the monotonic bias inherited from ordered pre-training data by shuffling frames at inference; this expedient, however, forfeits the temporal order that is indispensable both to temporal reasoning and to the disambiguation of visually similar yet temporally distinct states. Our approach, in contradistinction, preserves chronological order while capturing temporal context through test-time adaptation, and so eludes such shortcuts.

5.2 TEST-TIME ADAPTATION

Test-time adaptation methods afford parameter-efficient means of adapting VLMS without recourse to full fine-tuning. Prompt-based methods optimize model parameters at test time in light of the task context, though the mechanisms underlying them remain opaque (Zhou et al., 2022; Khattak et al., 2023; Ma et al., 2023a). Lim et al. (2025) conjectured that CLIP representations might sustain test-time adaptation, reasoning from the observation that CLIP encodes human-interpretable concepts unearthed through mechanistic interpretability (Bricken et al., 2023). CLIP-based similarity has likewise been enlisted as a reward signal, with parameters optimized at test time via reinforcement learning (Zhao et al., 2024). A further approach is test-time training (Sun et al., 2020), in which an adaptation module is updated at each timestep under a self-supervised loss. In contradistinction to meta-reinforcement learning methods (Finn et al., 2017; Bauer et al., 2023), which necessitate task-specific adaptation episodes during training, test-time training permits direct online adaptation during inference, obviating any need for task labels.

6 DISCUSSION, LIMITATIONS, AND FUTURE WORK

6.1 DISCUSSION

We have shown that test-time adaptation enables value functions for robotic manipulation to generalize across distribution shifts in task, environment, and embodiment. VITA, in addition, can distinguish expert from non-expert visual trajectories and effect zero-shot reward shaping, thereby opening the way to downstream applications in RL and imitation learning. By updating sequentially over a trajectory, VITA inscribes the accumulated history into its parameters, and so apprehends

temporal and semantic context more effectively than do either the CLIP-based baselines or in-context learning with autoregressive VLMs. That VITA outperforms CLIP-GRU on both expert–non-expert discrimination and offline RL indicates that, for zero-shot value function estimation, implicit memory realized through sequential test-time updates generalizes more effectively than the explicit encoding of temporal history within recurrent hidden states. Pre-trained autoregressive VLMs (GVL), for all their generalization capabilities, are not trained expressly for progress estimation—a task that demands temporal reasoning over visual trajectories. Within our setup, the inference-time overhead of per-timestep adaptation proves negligible, since only a lightweight adaptation module is updated at test time, which leaves real-time applicability unimpaired.

6.2 LIMITATIONS AND FUTURE WORK

Although, in our experiments, test-time adaptation improves the generalization of zero-shot progress estimation across unseen tasks and environments, it may yet encounter difficulties in settings marked by high execution variability or protracted durations. And while the adaptation overhead remains negligible by virtue of the lightweight adaptation module, updating a value function estimator at every timestep may prove potentially unsafe during deployment, thereby constraining applicability in real-time scenarios. In future work, we intend to explore alternative approaches to the test-time adaptation of vision–language models, particularly within the context of training agents inside world models. The application of VITA to real-time closed-loop control settings and to complex RL environments likewise remains an auspicious direction for future work. Being agnostic to the underlying vision–language encoder, VITA could augment the generalization and temporal reasoning of any domain-specific pre-trained multimodal representation by way of meta-learned test-time adaptation. Finally, although we furnish empirical evidence that dissimilarity-based sampling attenuates shortcut learning, a theoretical analysis of how diversity-based sampling governs shortcut learning remains an inviting avenue; this, however, lies beyond the compass of our core contribution, which introduces a value function estimator that meta-learns a test-time adaptation mechanism so as to improve zero-shot performance and temporal reasoning.

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A TRAINING DATASET

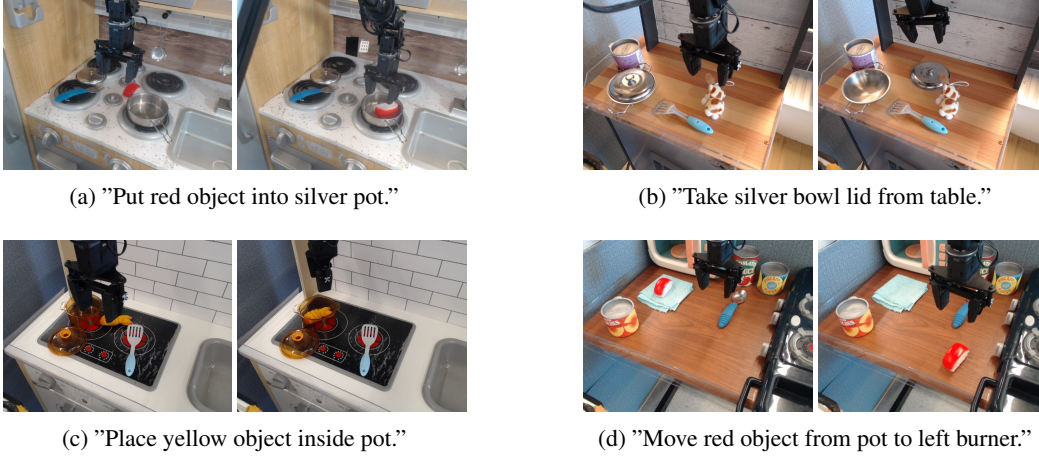


Figure 4: Each subfigure presents the start and end frames of an expert demonstration employed for training, accompanied by its natural-language task description. The demonstrations were collected across four distinct ToyKitchen environments.

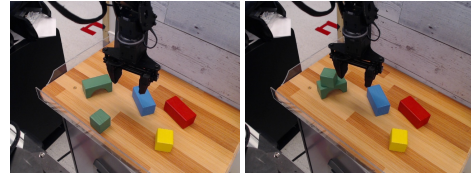
B EVALUATION DATASETS

Table 3: Descriptions of the datasets, itemized by task type, environment, and embodiment.

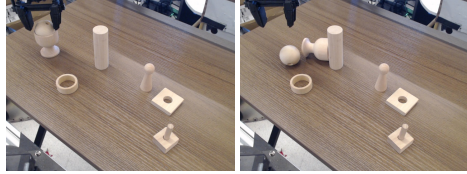
Dataset	Task Type	Environment	Embodiment
tk_pnp	pick-and-place	toy kitchen	WidowX 250
lm_pnp	pick-and-place	laundry machine	WidowX 250
td_fold	fold cloth	tabletop (dark wood)	WidowX 250
ft_fold	fold cloth	folding table	WidowX 250
rd_fold	fold cloth	robot desk	WidowX 250
ms_sweep	sweep	folding table (tray)	WidowX 250
dt_tk_pnp	pick-and-place	toy kitchen	DeepThought
dt_tk_stack	stack blocks	toy kitchen	DeepThought
dt_ft_stack	stack blocks	folding table	DeepThought
dt_rd_pnp	pick-and-place	robot desk (drawer)	DeepThought



(a) dt_tk_ms: "Move yellow object from burner to center."



(b) dt_tk_stack: "Place green cube on top of arch."

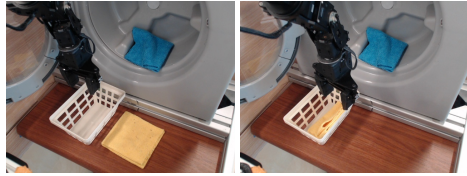


(c) dt_ft_stack: "Move egg to table."



(d) dt_rd_pnp: "Put orange object in drawer."

Figure 5: Each subfigure presents the start and end frames of an evaluation trajectory under embodiment shift, accompanied by its natural-language task description. The top row portrays tasks conducted in the same environment (ToyKitchen) with a different robot (DeepThought), whereas the bottom row encompasses tasks that additionally implicate novel environments.



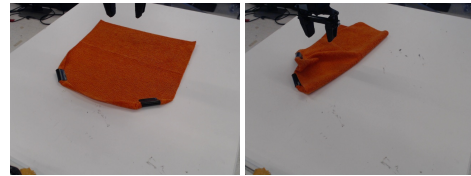
(a) lm_pnp: "Place blue cloth inside washer."



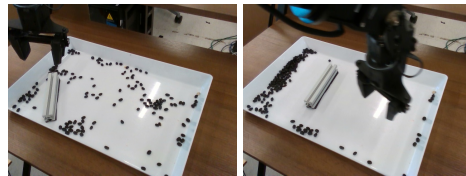
(b) td_fold: "Fold cloth bottom right to top left."



(c) ft_fcnp: "Fold clothes from left to right."



(d) rd_fold: "Fold cloth bottom right to top left."



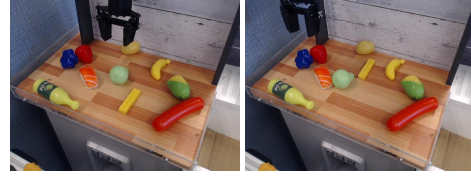
(e) ms_ft_sweep: "Sweep into pile."

Figure 6: Each subfigure presents the start and end frames of an evaluation trajectory under environment shift, accompanied by its natural-language task description. All tasks are executed with the same robot embodiment across environments that differ both visually and structurally.

C SCRIPTED DATASET



(a) `sc_pnp_obj`: "Pick up an object and place it."



(b) `sc_pnp_robj`: "Pick up a rigid object and place it."



(c) `sc_pnp_stoy`: "Pick up a soft object and place it."



(d) `sc_pnp_uten`: "Pick up a kitchen utensil and place it."



(e) `sc_pnp_cloth`: "Move the cloth."

Figure 7: Each subfigure presents the start and end frames of a scripted evaluation trajectory collected in the ToyKitchen environment with the WidowX 250 robot. All tasks involve pick-and-place behavior across varying object categories. We resort to generic task descriptions on account of the absence of publicly available annotations.

D MODEL ARCHITECTURE

D.1 MODEL AND TRAINING DETAILS

We adopt, as a frozen backbone, the OpenCLIP ViT-B/32 encoder pre-trained on the OpenAI dataset. Each visual observation and each task description is encoded, respectively, by the CLIP vision and text encoders so as to yield its representation. We elected a joint (concatenation-based) representation in preference to an element-wise product, a choice dictated by its superior validation performance. The adaptation module f_{adapt} takes the form of a two-layer residual MLP with GELU activation and a projection dimension $d' = 64$. The model is trained by means of the AdamW optimizer, with a learning rate of 1×10^{-4} , weight decay of 1×10^{-4} , and a cosine learning-rate schedule endowed with a 10% warmup. We employ a batch size of 32 and pad all trajectories to a maximum length of 120 frames (matching the longest sequence in the training set). The weighting coefficient λ_{self} governing the self-supervised loss within the total training objective is fixed at 0.5, again on the basis of validation performance. We train for 5 epochs, since the validation VOC characteristically plateaus early and prolonged training confers no further improvement. For dissimilarity-based sampling, we set the stride to $s = 1$, the sub-trajectory length to $w_{\text{tr}} = 8$, and the number of selected windows to $k = 8$, except where otherwise specified. All experiments were conducted on NVIDIA RTX 6000 Ada Generation GPUs.

D.2 TEST-TIME TRAINING HYPERPARAMETERS

At inference, we adapt the temporal adaptation module f_{adapt} alone, retaining the same projection dimension ($d' = 64$) as in training. We perform a single gradient step ($t_{\text{ep}} = 1$) with a learning rate

Table 4: The bearing of the test-time learning rate η and the number of adaptation steps t_{ep} upon the performance of VITA on the validation set, as measured by VOC.

η	$t_{\text{ep}} = 1$	$t_{\text{ep}} = 5$	$t_{\text{ep}} = 10$
0.001	0.225	0.345	0.453
0.01	0.454	0.744	0.781
0.05	0.746	0.670	0.515
0.1	0.783	0.512	0.369
1.0	0.299	0.183	0.181
5.0	0.169	0.182	0.157

$\eta = 0.1$, a configuration culled from a sweep over learning rates $\{5.0, 1.0, 0.1, 0.05, 0.01, 0.001\}$ and adaptation steps $\{1, 5, 10\}$ on the strength of the validation performance reported in Table 4. The results reveal that a single update with a larger learning rate ($\eta = 0.1$, $t_{\text{ep}} = 1$) and multiple updates with a smaller learning rate ($\eta = 0.01$, $t_{\text{ep}} = 10$) alike attain comparable validation performance under test-time training. We favor the former, inasmuch as it demands fewer adaptation steps while yielding marginally superior results.

D.3 BASELINES

VITA employs $t_{\text{ep}} = 1$, a projection dimension $d' = 64$, and a learning rate η of 0.1, all selected via the hyperparameter tuning described previously. CLIP-FT conforms to the architecture of VITA but forgoes the adaptation module and the self-supervised loss; it comprises a frozen CLIP encoder followed by a linear projection and a two-layer MLP that predicts task progress. The model is trained by standard supervised regression, without meta-learning or test-time adaptation, and, so as to augment its expressivity, it is endowed with an $8\times$ larger projection matrix and $10\times$ more training steps than our method (Lin et al., 2022). For VLM-RM, we adopt a baseline prompt that describes the environment. CLIP-GRU likewise follows the architecture of VITA (a frozen CLIP encoder and regression head h), but supplants the test-time adaptation module f_{adapt} with a Gated Recurrent Unit (GRU) module that encodes temporal history explicitly through recurrent hidden states; the GRU module consists of a single layer with hidden size 64, whose final hidden state is relayed to the regression head h . For both GVL-0S and GVL-1S, we employ the latest release of Gemini 1.5 Pro, gemini-1.5-pro-latest (Gemini Team et al., 2024), whereas the original GVL implementation drew upon an earlier release, gemini-1.5-pro. We further evaluated GVL with the open-source autoregressive VLM Qwen-VL 2.5 (Yang et al., 2024), which proved unable to surmount the temporal bias notwithstanding frame shuffling, while GPT-4o (Hurst et al., 2024) persistently declined to produce scalar progress estimates in our setup. All evaluations abided by the prompt template introduced in the original GVL work (Ma et al., 2024).

E OFFLINE RL SETUP

E.1 TRAINING CONFIGURATION

We employ Implicit Q-Learning (IQL) under the following settings, which are held fixed across all tasks. The expectile regression is set to $\tau = 0.7$, with an advantage-weighting temperature of 3.0 and a clipping threshold of 100.0. The actor and critic assume learning rates of 1×10^{-4} and 3×10^{-4} , respectively, together with a batch size of 256. Policies are trained for 100,000 gradient steps, with evaluation carried out every 10,000 steps over 20 rollouts per task (horizon 150). The implementation follows d3rlpy, with the configuration: expectile=0.7, weight-temp=3.0, max-weight=100.0, actor_lr=1e-4, critic_lr=3e-4, batch-size=256.

E.2 EVALUATION PROTOCOL

In accordance with Agarwal et al. (Agarwal et al., 2021), we report the interquartile mean (IQM) of the returns across all task-seed pairs. The IQM, being the mean of the middle 50% of the scores, curbs sensitivity to outliers more effectively than either the mean or the median. To secure statistical

robustness, we estimate 95% confidence intervals by stratified bootstrap with 10,000 resamples, stratified by task. This procedure is applied uniformly across all methods in the Meta-World MT10 experiments.

F ABLATION STUDIES

F.1 EFFECT OF DISSIMILARITY-BASED SAMPLING ON DISCRIMINATIVE PERFORMANCE

Table 5: Disc@VOC results across the scripted datasets under divergent training-time sampling strategies for test-time adaptation. A checkmark denotes successful discrimination between expert and suboptimal trajectories. The sampling methods **FT** (full-trajectory), **Rand** (random sampling), and **Diss** (dissimilarity-based sampling) are each parameterized by (w_{tr}, k) .

Dataset	FT (8,-)	Rand (8,8)	Diss (8,4)	Diss (4,8)	Diss (8,8)
sc_pnp_obj		✓	✓		✓
sc_pnp_robj	✓	✓	✓	✓	✓
sc_pnp_stoy			✓		✓
sc_pnp_uten	✓	✓	✓	✓	✓
sc_pnp_cloth		✓	✓	✓	✓
Avg. Disc@VOC	0.20	0.8	1.0	0.60	1.0

We validate the efficacy of the proposed dissimilarity-based sampling empirically, contrasting the impact of divergent sub-trajectory sampling strategies; more precisely, we examine how the sampling strategy adopted during training bears upon the ability of our model to discriminate expert from suboptimal trajectories. In Table 5, we compute Disc@VOC for 5 scripted datasets, thereby capturing whether a given model ascribes a higher average VOC to expert trajectories (tk_pnp) than to each evaluation dataset. We juxtapose three strategies: (1) full-trajectory sampling, which applies adaptation over all overlapping sub-trajectories of length w_{tr} with stride s , in keeping with prior work (Sun et al., 2024); (2) random sampling, which draws k sub-trajectories of length w_{tr} uniformly per video; and (3) our proposed dissimilarity-based sampling, which assembles a candidate set of sub-trajectories of length w_{tr} with stride s and thereupon selects the k sub-trajectories that maximize pairwise dissimilarity within the representation space.

Our results demonstrate that dissimilarity-based sampling with $w_{tr} = 8$ and $k = 8$ yields flawless discriminative performance. Sampling fewer windows ($k = 4$) of size $w_{tr} = 8$ issues in the same performance, whereas curtailing the sub-trajectory length to $w_{tr} = 4$ diminishes effectiveness. The full-trajectory baseline fares the worst, corroborating that adapting over all consecutive windows can engender overfitting to global temporal cues. Random sampling outstrips full-trajectory sampling, yet remains less dependable than dissimilarity-based selection. Taken together, these findings accentuate the importance of both sub-trajectory diversity and semantic locality in enabling discriminative test-time adaptation.

F.2 EFFECT OF TEMPORAL MEMORY ON VALUE FUNCTION ESTIMATION

We evaluate four adaptation strategies: (1) TTT-TR, which enacts a single gradient update computed over the full trajectory (offline); in performing a batched update that averages the loss across all frames, TTT-TR thereby discards temporal order; (2) TTT-RS, which resets the adaptation module at every step and updates it with the current visual observation alone (memoryless), and so precludes the propagation of any temporal information; (3) TTT-EX, which likewise resets the adaptation module at every step but updates it with a local window of recent frames matching the window size used during meta-training ($w_{tr} = 8$); in averaging gradients over this window, TTT-EX forfeits the temporal ordering within it; (4) VITA (OURS), which updates the adaptation module incrementally at every timestep (implicit memory), forming an implicit memory through the sequential accumulation of temporal history within the adapter parameters. We conducted identical test-time hyperparameter sweeps for all variants, as described in Appendix D, and found that VITA, TTT-RS, and TTT-TR attain optimal performance with a learning rate of 0.1 and a single adaptation step, whereas TTT-EX performs best with a learning rate of 1.0 and one adaptation step.

Table 6: Validation VOC scores for progress estimation under an assortment of distribution shifts. ID = In-Distribution, ES = Environment Shift, EM = Embodiment Shift, ES+EM = Conjoint Environment and Embodiment Shift.

Shift	Dataset	TTT-TR	TTT-RS	TTT-EX	VITA
ID	tk_pnp	0.1917	0.1918	0.1957	0.7822
ES	lm_pnp	0.1843	0.1854	0.1826	0.7246
	td_fold	0.1402	0.1393	0.1322	0.7085
	ft_fold	0.1302	0.1311	0.1683	0.6583
	rd_fold	0.1161	0.1172	0.1154	0.6056
	ms_sweep	-0.0225	-0.0191	0.0102	0.4898
EM	dt_tk_pnp	0.2123	0.2118	0.2069	0.8203
	dt_tk_stack	0.0833	0.0830	0.0747	0.7081
ES+EM	dt_ft_stack	0.0558	0.0537	0.0439	0.6979
	dt_rd_pnp	0.1665	0.1660	0.1559	0.6951

The results in Table 6 reveal that TTT-TR, TTT-RS, and TTT-EX yield comparable performance—an indication that batched updates, whether computed over a short explicit window (TTT-EX) or over the full trajectory (TTT-TR), perform on a par with the memoryless baseline (TTT-RS). VITA, by contrast, consistently outperforms all three variants, thereby underscoring the importance of sequential, per-timestep updates for effective value function estimation in real-world robotic manipulation tasks. Under implicit memory (VITA), updating over a local window of frames would occasion the most recent observations within that window being incorporated twice—explicitly through the window and implicitly through the accumulated parameters—while updating less frequently (that is, non-sequentially) would precipitate a loss of information, inasmuch as intermediate frames are omitted. The implicit-memory test-time training paradigm accordingly requires sequential, per-timestep updates without reset.

G COMPLEXITY ANALYSIS OF DISSIMILARITY-BASED SAMPLING

Let $\mathcal{W}$ denote the candidate set of size $N_c = |\mathcal{W}|$. In order to approximate diverse subset selection without incurring the exponential cost of enumerating all $\binom{N_c}{b}$ subsets, as would be entailed by Eq. 4, we adopt a scoring-based heuristic that assigns to each candidate window a diversity score equal to the sum of its pairwise distances from all others (that is, the row sum of the distance matrix). Let the number of candidate windows be $N_c = T - w_{tr} + 1$, each represented by a flattened feature vector of dimension $m = w_{tr} \cdot d$, where d is the dimension of each multimodal representation z_t . We compute the full pairwise distance matrix $D \in \mathbb{R}^{N_c \times N_c}$ on the GPU by means of batched `cdist`, an operation that demands $\mathcal{O}(N_c^2 \cdot w_{tr} \cdot d)$ computations.

In our experiments, the training trajectories are short (of mean length 69 and maximum length 120). With a batch size of 32, a feature dimension $d = 1024$ (the CLIP-based multimodal representation), and a window length $w_{tr} = 8$, the worst-case computational cost amounts to:

$$32 \cdot N_c^2 \cdot w_{tr} \cdot d = 32 \cdot 114^2 \cdot 8 \cdot 1024 \approx 340 \text{ MFLOPs.}$$

This overhead is negligible relative to the cost of a single forward and backward pass through our model, all the while promoting the selection of diverse windows.